

Reflection Paper

1. What Did I Implement?

I used the cleaned car dataset from Assignment Three to build two regression models for predicting car prices. I selected Price as the target variable and used the remaining features, except LogPrice, as predictors. The dataset was split into 80% for training and 20% for testing using `random_state=42`. I trained both a Linear Regression model and a Random Forest Regressor, then evaluated them using R^2 , MAE, MSE, and RMSE.

2. Comparison of Models

In the sanity check, the Random Forest prediction was closer to the actual car price than the Linear Regression prediction. However, Linear Regression achieved a higher overall R^2 score. This shows that a single prediction does not always represent the overall performance of a model.

3. Understanding Random Forest

Random Forest is an ensemble learning algorithm that combines many decision trees. Each tree makes its own prediction, and the final prediction is the average of all trees. This approach usually improves stability and reduces the effect of individual tree errors.

4. Metrics Discussion

In this project, Linear Regression achieved a higher R^2 score, while Random Forest produced a more accurate prediction for the selected sanity check sample. R^2 measures performance across the entire test dataset, whereas the sanity check evaluates only one observation. For this reason, a model may predict one sample very well but still have lower overall performance.

5. Why Did Random Forest Have a Lower R^2 but Perform Better in the Sanity Check?

Random Forest predicted the selected car more accurately because it captured the local pattern for that specific example. However, R^2 is calculated using all test samples. If Random Forest makes larger errors on many other cars, its overall R^2 becomes lower than Linear Regression. Therefore, the sanity check should not be used alone to judge model quality.

6. My Findings

I prefer the Linear Regression model for this dataset because it achieved a higher R^2 score and better overall performance. Although Random Forest was closer to the actual value in the sanity check, Linear Regression generalized better across the full test dataset. This demonstrates the importance of evaluating models using multiple metrics instead of relying on a single prediction.